

- *Analog velocity measurement using a tachometer*

A tachometer is a D.C. machine which provides a D.C. tension proportional to the rotation velocity of its armature. Its sensitivity is generally of the order of a few volts per 1000 rpm.

Its use is thus reserved to the measurement of relatively high velocities and it has to be mounted on the actuator axis.

It has good linearity properties ($\simeq 10^{-3}$), but its use remains limited for cost reasons.

It has to be associated with an analog-to-digital converter in order to provide a measurement into suitable digital form for further treatment by the control unit.

2.14 The robot manipulator ASEA-IRb-6

In order to illustrate the preceding considerations, let us consider as an example the robot manipulator ASEA-IRb-6 represented by figure 2.14.1 It possesses a mechanical architecture in which the main kinematic chain is of RRRRR type. Its wrist possesses only 2 DOF in the standard version. The five degrees of freedom correspond to:

1. the waist rotation ψ producing arm sweep,
2. the shoulder rotation ϕ producing upper-arm motion,,
3. the elbow rotation α producing lower-arm motion,,
4. the wrist bending t ,
5. the wrist twist v .

The manufacturer has opted for the choice of electrical motorization (servo-controlled D.C. motors) centralized in the basis. Therefore, the different DOF are actuated through secondary kinematic chains parallel to the main one.

The displacement ranges of the successive DOF and the associated velocities are:

waist rotation	340°	95 °/s
shoulder rotation	±40°	radial velocity of 0.75m/s
elbow rotation	±25°	vertical velocity of 1.1m/s
wrist bending	±90°	115°/s
wrist twist	±180°	195°/s

The corresponding workspace is represented hereafter in perspective view on one hand, and through a vertical plane cut on the other hand. The manufacturer does not specify the accuracy of the machine, but its repeatability is 0.2mm. A certain number of mechanical characteristics are given by the technical sheet provided at the end of the chapter.

Mechanical design

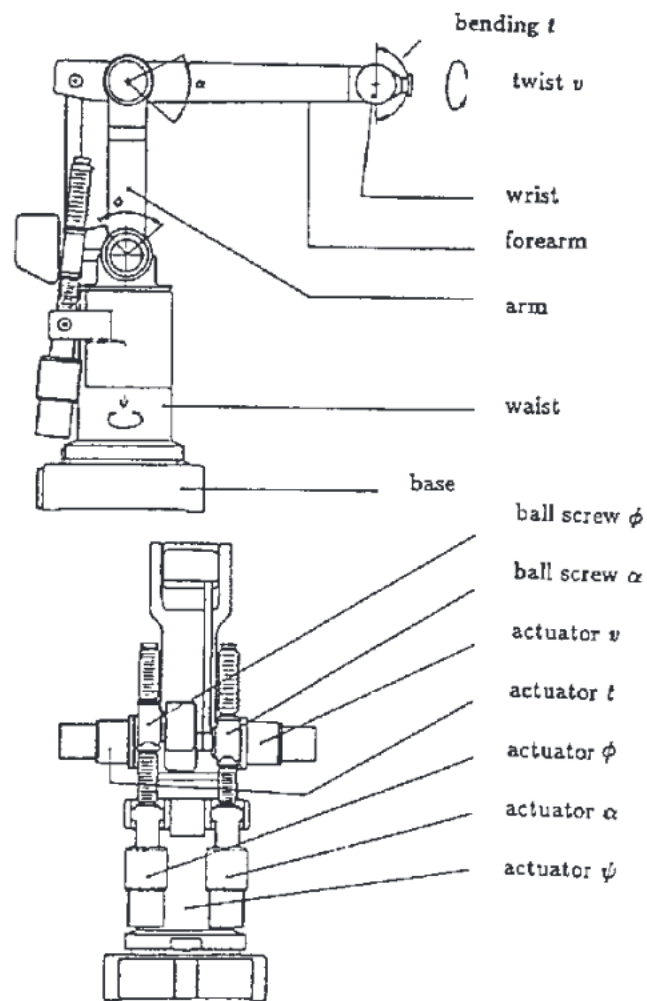


Figure 2.14.1: Mechanical design of the robot ASEA IRb-6

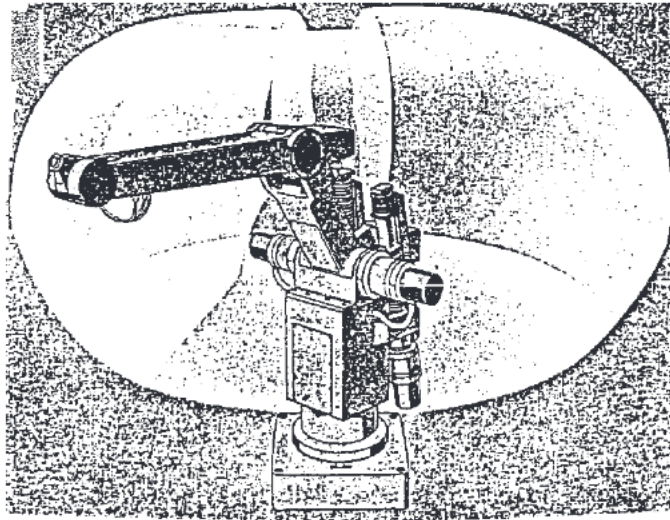


Figure 2.14.2: Workspace of the ASEA robot IRb-6: perspective representation

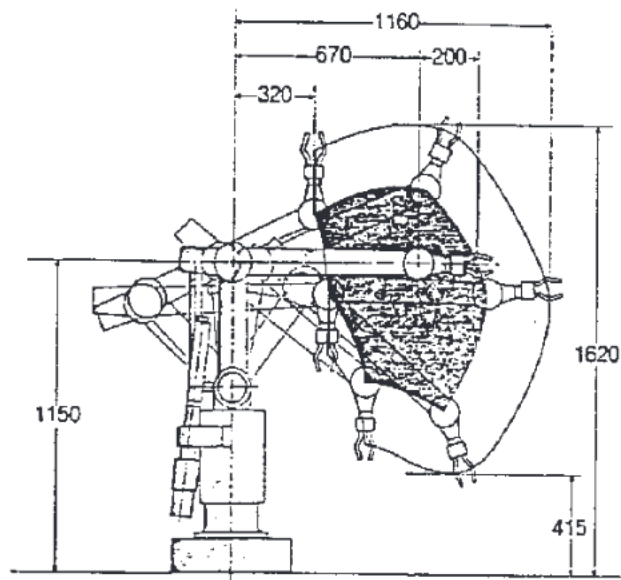


Figure 2.14.3: Workspace of the ASEA robot IRb-6: vertical cut representation

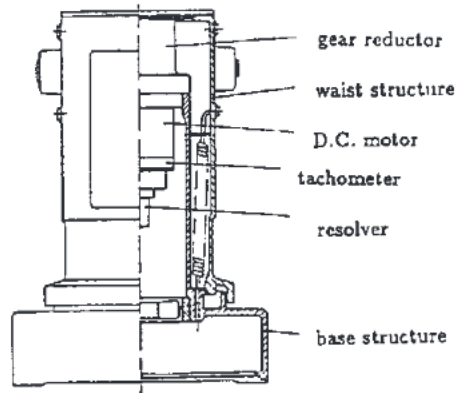


Figure 2.14.4: Motor unit for waist rotation



Figure 2.14.5: Four bar mechanisms for motion transmission

Figure 2.14.4 displays the mechanical design of the first DOF: the whole motor unit is rigidly attached to the base of the manipulator and generates rotary motion of the base through a gear reductor. The tachometer provides velocity information and the resolver, positioning information. The motor units for the DOFs ϕ and α (shoulder and elbow rotations) are also localized in the base, the motion transmission being achieved through four bar linkages which constitute the most elementary closed-loop mechanisms (figure 2.14.5). Figure 2.14.6 displays the motor unit of the shoulder: the actuator rotary motion is transformed into translation motion through a *ball screw* whose nut acts on a lever hinged to the arm. In this way, a rotation-rotation transformation of motion with a high reduction rate is obtained while allowing us to locate the motor unit of the shoulder in the base of the manipulator. The ball screw is a very efficient implementation of the screw joint (figure 2.14.7). It is made of a helicoidal screw whose thread is a rolling path for the balls and a nut containing them. The balls are recirculated from one end of the nut to the other through external ducts.

The motor unit of the elbow is designed on the same principle (figure 2.14.8). An additional mechanical loop is however needed to transmit the motion to the end of the arm.

The motor units for wrist bending and twist are also located in the base of the structure, and the motion transmission from the base to the wrist is obtained through a parallelogram mechanism, as shown by figure 2.14.9 The transmission is made of three quaternary links having the form

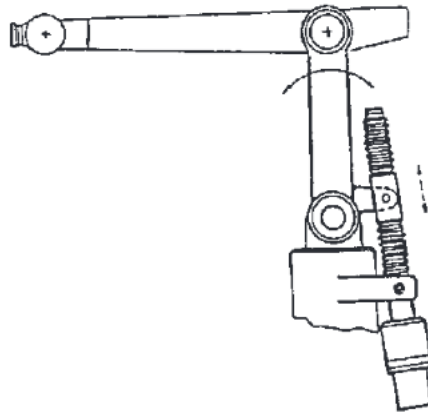


Figure 2.14.6: Motor and transmission unit for shoulder rotation

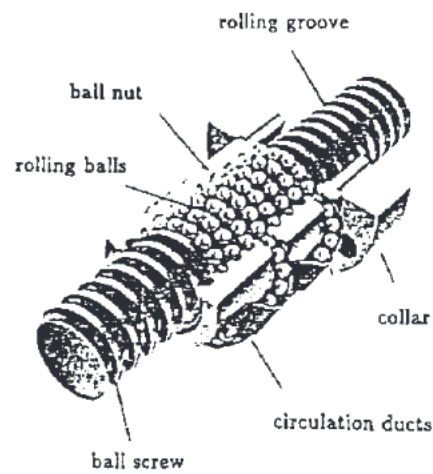


Figure 2.14.7: Ball screw implementation of the screw joint

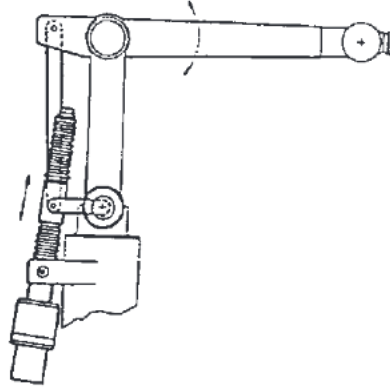


Figure 2.14.8: Motor and transmission unit for elbow rotation

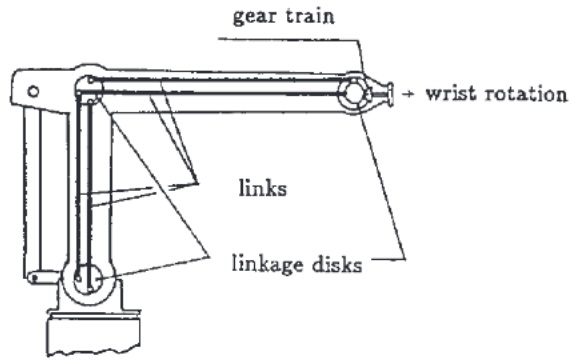


Figure 2.14.9: Transmission unit for wrist bending

of disks and located at motor, elbow hinge and wrist levels, respectively. They are connected together through parallel links which are hinged on the disks at points distant from 90° . Two similar linkages going through the main structure are necessary to transmit both bending and twist DOF (figure 2.14.9). For the wrist DOFs, the speed reduction is achieved first through gear reducers of *harmonic drive* type. Harmonic drives are highly performing reducers based on a rather simple but clever principle. They are made of three parts:

- the output element is a *rigid planetary* of annular shape with internal teeth ;
- the input element is a *elliptic wave generator* mounted on a roller bearing of same shape;
- the intermediary element is a *deformable satellite*, with a number of teeth slightly inferior (generally, 2) to that of the planetary.

One can show that, if Z_p and Z_s are respectively the numbers of teeth on the planetary and on the satellite, the reduction ratio is given by the formula

$$N = \frac{Z_p}{Z_p - Z_s} \quad (2.14.1)$$

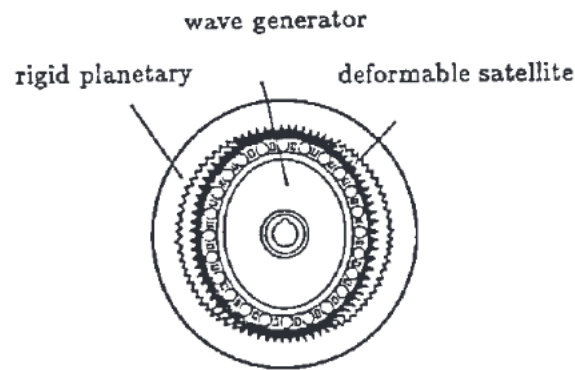


Figure 2.14.10: Reductor of harmonic drive type

In this way, reduction ratios as high as 320 can be obtained in a single reductor stage. The number of teeth participating in the contact is rather high (about 10 %); this is favorable to wear and backlash reduction and increases at the same time the mechanical stiffness.

2.15 Technical sheets of some industrial robot manipulators

Starting from the technical data provided by manufacturers, it is generally possible to summarize the characteristics of a specific manipulator in a technical sheet made on the same model as the ones given hereafter. However, it is not always possible to gather all the data which describe completely the manipulator under consideration.

2.15.1 Industrial robot ASEA IRb-6/2

KINEMATIC ARCHITECTURE: 5 R			
DEGREES OF FREEDOM:			
q_1	R	340°	90°/s
q_2	R	80°	1m/s
q_3	R	50°	1.35m/s
q_4	R	180°	90°/s
q_5	R	360°	150°/s
MOTORIZATION: D.C. servomotors			
MECHANICAL PERFORMANCES: capacity: 6kg max. linear velocity: r direction 0.75 m/s; z direction 1.1m/s. repeatability: 0.2 mm accuracy: not specified			
EFFECTOR: various grippers adapted to specific applications			
SENSORS: resolvers + tachometers			
CONTROL UNIT: central unit with 16K ram			
PROGRAMMING MODE: - teach box - computer link			
APPLICATIONS: - arc welding - assembly - inspection - machining - surface coating - glueing			
Industrial robot ASEA IRb-6/2: technical data			

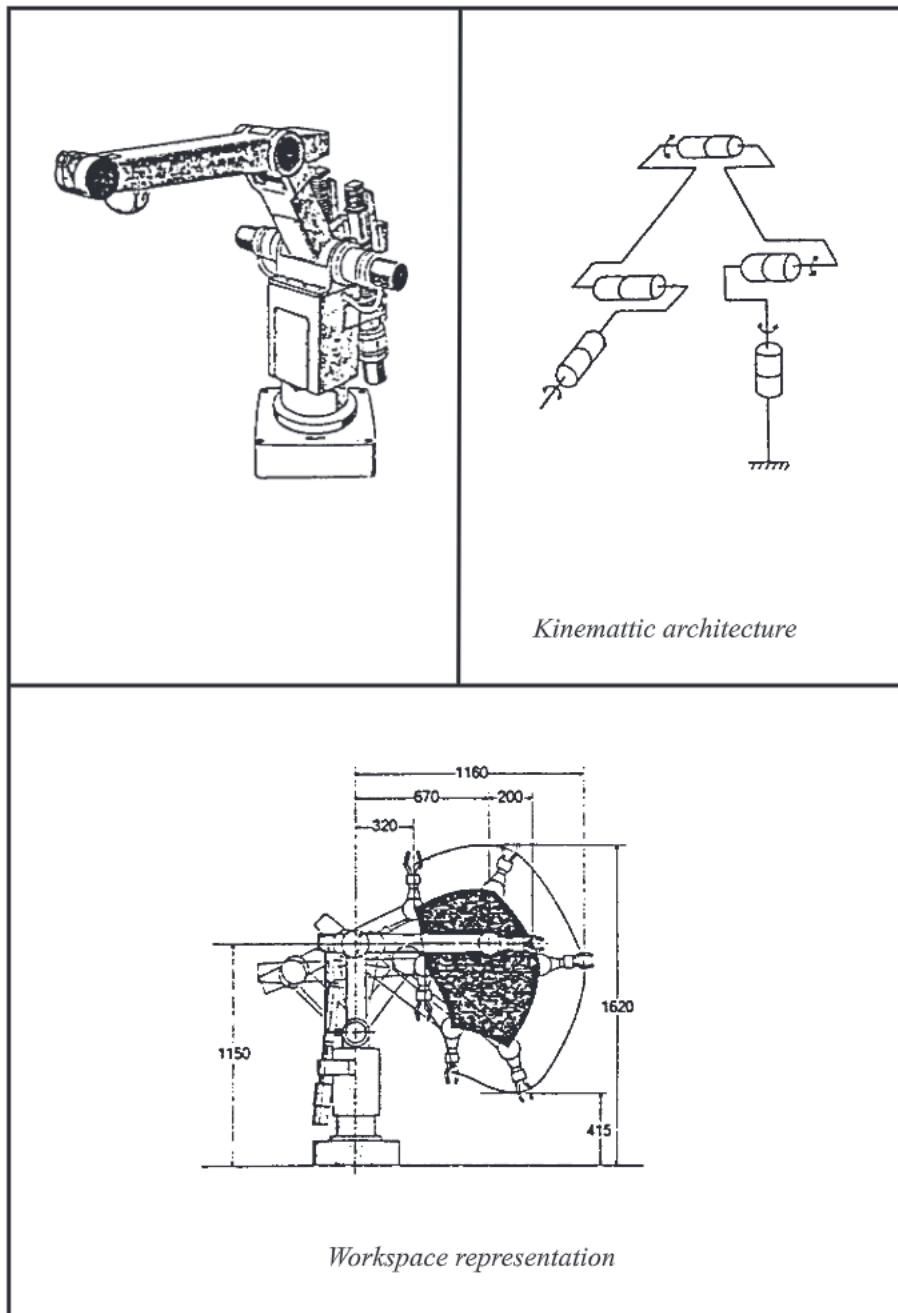


Figure 2.15.1: Industrial robot ASEA IRb-6/2: geometric configuration

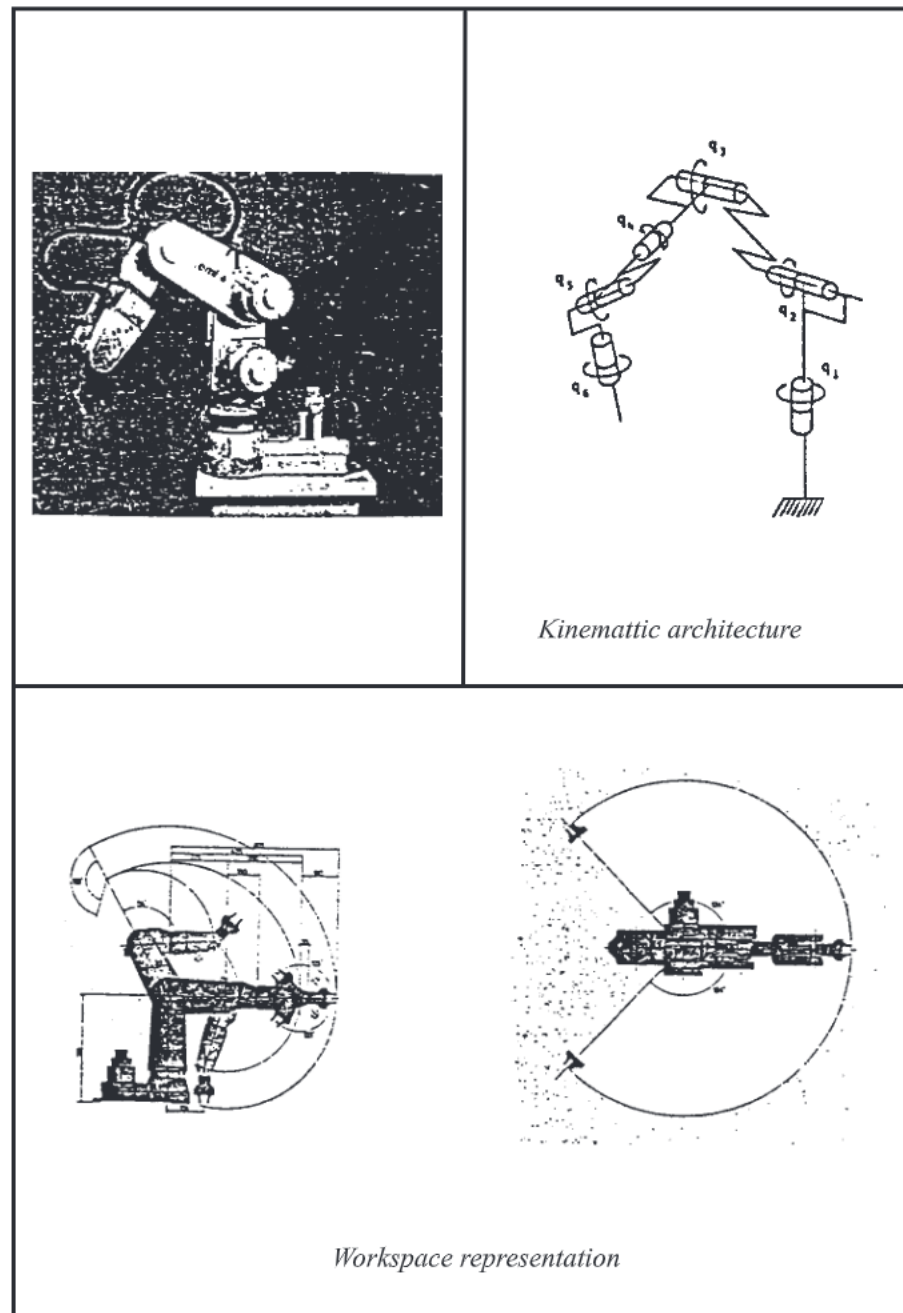


Figure 2.15.2: Industrial robot SCEMI 6P-01: geometric configuration